Project Design Ideation Phase

Empathize & Discover — Empathy Map Canvas

Date	09 March 2026
Team ID	698c67ecb31ff1dfca225748
Project Name	PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning
Team Members	Akshat Bhati, Shubhadipta sarkar, Haridharshan B, Madan M
Maximum Marks	4 Marks

Empathy Map Canvas

User Persona: Ravi Kumar — 34-year-old smallholder farmer, Nashik district, Maharashtra. Grows tomatoes, potatoes, and chillies across 3 acres. Owns a basic Android smartphone. Family income entirely dependent on seasonal crop yield.

WHAT DOES THE USER THINK & FEEL?

What really counts • Major preoccupations • Worries & aspirations

- "My family's entire livelihood depends on this harvest — one bad season could push us into serious debt."
- Constantly anxious about whether leaf yellowing or dark spots are normal environmental stress or the onset of a serious disease — the uncertainty is paralysing
- Deeply worried about applying the wrong pesticide after a misdiagnosis, wasting money and causing chemical damage to crops
- Aspires to be a modern, tech-savvy farmer; wants peers and family to see him confidently adopting new digital tools
- Believes early disease detection is the single most critical factor in saving a crop and securing the season's income
- Worries that by the time expert confirmation arrives, the disease will have spread irreversibly across the field
- Hopes PlantCare AI can become his always-available 'expert in his pocket' — accessible even at 5 AM when symptoms first appear

WHAT DOES THE USER HEAR?

What friends say • What influencers say • What market offers

- Neighbours: "I lost 40% of my tomatoes because I didn't catch the blight early enough"
- Village co-op officer giving generic advice: "Just spray neem oil on everything" — not specific enough
- Radio agri-programs warning about disease outbreaks spreading in nearby districts this growing season
- WhatsApp farming groups with blurry leaf photos and conflicting, unqualified diagnoses from fellow farmers

WHAT DOES THE USER SEE?

Environment • Friends • What the market offers

- His own fields showing yellow spots, brown patches, and unusual leaf curling that worry him every morning
- Neighbouring farms where undetected disease spread rapidly and destroyed entire crop plots within days
- Other farmers in the cooperative using smartphones for weather apps — digital adoption visibly growing around him

• Input dealers recommending expensive broad-spectrum pesticides regardless of actual disease — profit-driven advice
• Fellow farmers praising a 'leaf photo app' that correctly identified tomato late blight for them last season
• All India Radio news segments about AI tools being introduced in modern precision agriculture across India

• Pesticide shop shelves packed with hundreds of products with technical chemical names he cannot decipher
• Agricultural extension officer visiting the area only once every 2–3 months — rarely available during urgent outbreaks
• Media coverage of AI-powered precision farming — aspirational and motivating but feels distant from his daily reality
• Wife and family members asking "is the crop okay?" daily — amplifying stress and his sense of sole responsibility

WHAT DOES THE USER SAY & DO?

Attitude in public • Appearance • Behaviour towards others

• Tells neighbours: "I think my tomatoes have some kind of spot disease but I am not sure exactly what it is"
• Walks through his field every morning, manually inspecting leaves row by row, crouching close to examine each plant
• Takes blurry close-up photos of diseased leaves on his Android phone and posts them in WhatsApp farming groups seeking any opinion
• Visits the local pesticide/input shop, describes symptoms verbally, and buys whatever the shopkeeper recommends
• Sometimes applies 2–3 different pesticides simultaneously 'just in case' — an expensive and risky trial-and-error method
• Delays calling the agricultural extension officer because the officer is hard to reach and scheduled visits are rare
• After using PlantCare AI: photographs diseased leaf → uploads via browser → reads the diagnosis aloud to wife → shows neighbours the result with visible pride and confidence
• Begins actively recommending PlantCare AI to fellow farmers in his cooperative after his first successful diagnosis

PAIN

Fears • Frustrations • Obstacles

• FEAR: Total crop loss and financial ruin — no crop insurance or safety net to absorb the loss
• FRUSTRATION: No fast, reliable expert diagnosis available at nights, weekends, or during peak growing season
• FINANCIAL BARRIER: Private agronomist consultation costs ₹500–1500 per visit — unaffordable for recurring disease events
• KNOWLEDGE GAP: Cannot visually distinguish between 38 different plant diseases without specialist training
• TIME LOSS: Waiting days for the extension officer while disease spreads exponentially through the field
• WRONG TREATMENT: Buying incorrect pesticide after misdiagnosis — wasted money and possible crop damage from overuse

GAIN

Wants / Needs • Measures of success • Aspirations

• WANT: Instant, free, reliable plant disease diagnosis tool accessible on his basic Android phone browser
• NEED: Specific disease name + targeted treatment recommendation — not generic broad-spectrum pesticide advice
• ASPIRATION: To be a modern, data-driven farmer who catches problems early and protects maximum crop yield every season
• MEASURE OF SUCCESS: Upload a leaf photo and receive the correct disease diagnosis within 2 seconds — every single time
• SOCIAL GAIN: Being known as the farmer who uses AI — earning respect and becoming the go-to resource in his village
• FINANCIAL GAIN: Save 30–50% on pesticide costs by applying only the targeted treatment for the confirmed disease

• CONNECTIVITY: Inconsistent 4G in rural areas makes loading image-heavy web tools slow and unreliable	• EMOTIONAL GAIN: Confidence and peace of mind — diagnosis backed by AI trained on 87,000+ real plant disease images

User Persona Summary

Attribute	Detail
Persona	Ravi Kumar — Primary PlantCare AI End User (Farmer)
Age & Location	34 years old, Nashik, Maharashtra, India
Occupation	Smallholder farmer — 3 acres of tomatoes, potatoes, chillies
Tech Level	Basic — Android smartphone, WhatsApp, YouTube; comfortable with photo sharing apps
Income	~₹2.5–4 lakh/year; entirely dependent on seasonal crop health
Primary Goal	Detect plant disease early, get specific treatment advice, save maximum crop yield
Biggest Pain	Cannot access fast, expert-level diagnosis when symptoms first appear — time-critical window is lost
PlantCare AI Fit	Photograph leaf → upload via browser → instant 38-class AI diagnosis → disease name + recommended actions in under 2 seconds

Reference: https://www.mural.co/templates/empathy-map-canvas